AP PRECALCULUS · TOPIC 1.11

Equivalent Representations of Polynomial and Rational Expressions

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $f(x) = x^3 - 4x^2 - 5x$. Which list gives all of the zeros of f?

A. $-1,\ 0,\ 5$ **B.** $-5,\ 0,\ 1$

C. $-1,\ 5$ **D.** $0,\ 4,\ 5$

2 Let $r(x) = \dfrac{x^2 - x - 6}{x^2 - 9}$. What are the coordinates of the hole in the graph of r?

A. $(3, 0)$ **B.** $(3, 1)$

C. $\left(3, \dfrac{5}{6}\right)$ **D.** $\left(3, \dfrac{6}{5}\right)$

3 What is the quotient when $3x^3 - 2x^2 + x - 4$ is divided by $x - 1$?

A. $3x^2 - 2x + 1$ **B.** $3x^2 + x + 2$

C. $3x^2 - 5x + 6$ **D.** $3x^2 + x$

4 Which line is the slant asymptote of $r(x) = \dfrac{x^2 + 5x + 3}{x + 2}$?

A. $y = x + 5$ **B.** $y = x - 3$

C. $y = x + 2$ **D.** $y = x + 3$

5 What is the coefficient of x^2 in the expansion of $(x - 2)^4$?

A. -24 **B.** 6

C. 12 **D.** 24

6 Which statement about the graph of $r(x) = \dfrac{x^2 - 4}{x^2 - x - 2}$ is true?

A. Vertical asymptotes at $x = 2$ and $x = -1$

B. A hole at $x = 2$ and a vertical asymptote at $x = -1$

C. A hole at $x = -1$ and a vertical asymptote at $x = 2$

D. A hole at $x = -2$ and a vertical asymptote at $x = 1$

7 Which statement describes the end behavior of $f(x) = -2(x + 1)^2(x - 3)$?

A. $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$

B. $f(x) \to -\infty$ as $x \to \infty$ and as $x \to -\infty$

C. $f(x) \to -\infty$ as $x \to \infty$ and $f(x) \to \infty$ as $x \to -\infty$

D. $f(x) \to \infty$ as $x \to \infty$ and as $x \to -\infty$

8 Which is the expansion of $(2x - 1)^3$?

A. $8x^3 - 12x^2 + 6x - 1$ **B.** $2x^3 - 3x^2 + 3x - 1$

C. $8x^3 + 12x^2 + 6x + 1$ **D.** $8x^3 - 1$

AP PRECALCULUS · TOPIC 1.11 · SECTION 2

Equivalent Representations of Polynomial and Rational Expressions

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A drone-delivery company models its average cost per delivery, in dollars, by $A(n) = \dfrac{n^2 + 10n + 40}{n + 5}$, where $n \geq 1$ is the number of hundreds of deliveries made in a month.

(a) Use polynomial long division to write $A(n)$ as a quotient plus a remainder over $n + 5$. DOK3 2 pt

(b) Write the equation of the slant asymptote of the graph of A and interpret it in context. DOK3 2 pt

(c) Find the number of hundreds of deliveries for which the average cost is exactly 20 dollars. Give your answer to three decimal places. DOK3 2 pt

Question 10 No calculator · 6 points

Let $p(x) = x^3 - 3x^2 - 4x + 12$ and $r(x) = \dfrac{p(x)}{(x - 2)(x + 1)}$.

(a) Factor $p(x)$ completely and list its zeros. DOK3 2 pt

(b) Write $r(x)$ in simplified form and give the coordinates of the hole. DOK3 2 pt

(c) Use long division on the simplified form to find the slant asymptote of r. DOK3 2 pt